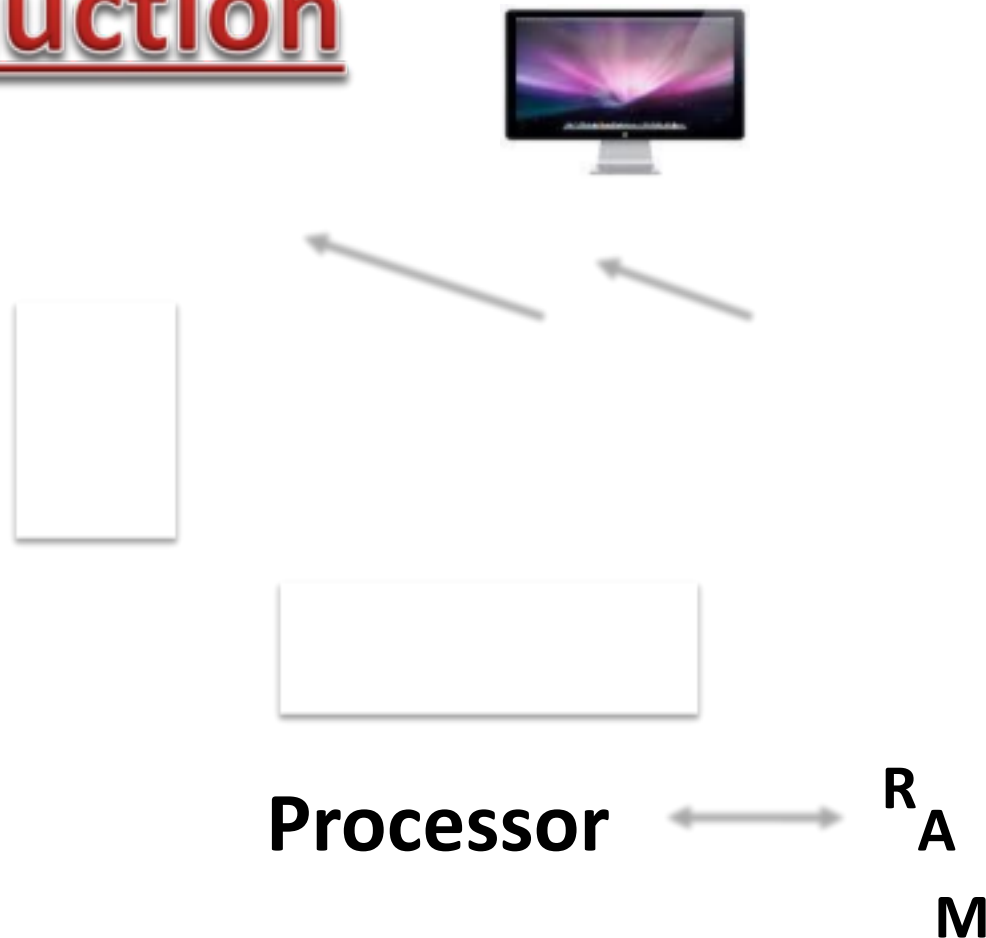


Interfacing Display with MPU ***without Peripheral Controller*** **Outline**

- Peripheral communications
- Display
- LED Display
 - Decoder
 - Generic model
 - Interfacing
- Character Display Monitor
 - Decoder

- Generic Model
- Graphics monitor
 - Why graphics? Vector Vs Raster
- Keyboard type and Interfaces

Introduction





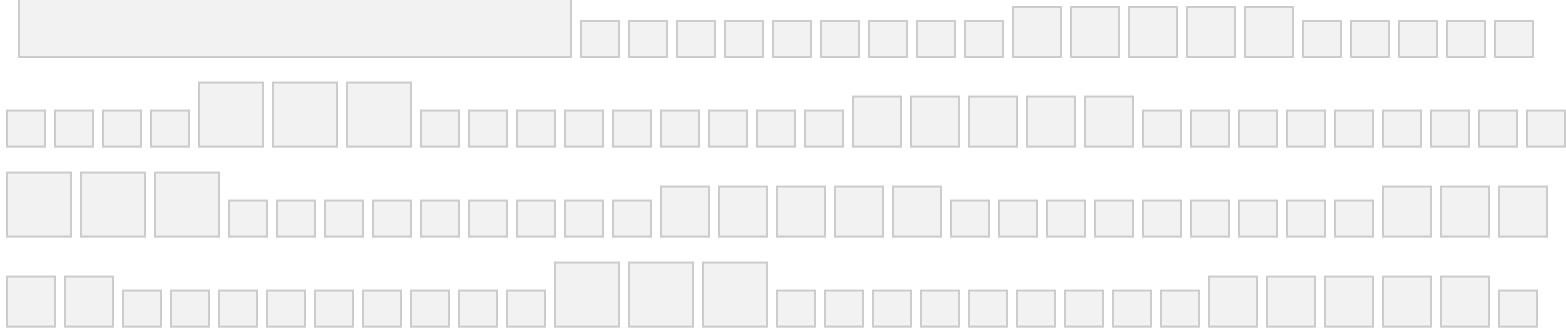
- Peripherals : HD monitor, 5.1 speaker
- **Interfaces : Intermediate Hardware**
 - Nvidia GPU card, Creative Sound Blaster card •
- Interfaces : Intermediate Software/Program –
 - Nvidia GPU driver , Sound Blaster Driver software
- Transmission Controller:

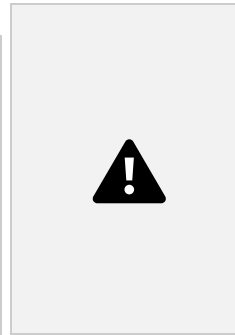


–

MPU control, Device Control (DMA)

- Type of IO mapping
 - **Peripheral (IN/Out)**, Memory mapped IO (LD/ST,MV)
- Format of communication
 - **Synchronous** (T & R sync with clock), **Asynchronous**
- Mode of Data Transfer
 - **Parallel**, Serial (UART)
- Condition for data transfer
 - **Uncond.**, Polling, Interrupt, **Ready signal**, Handshake

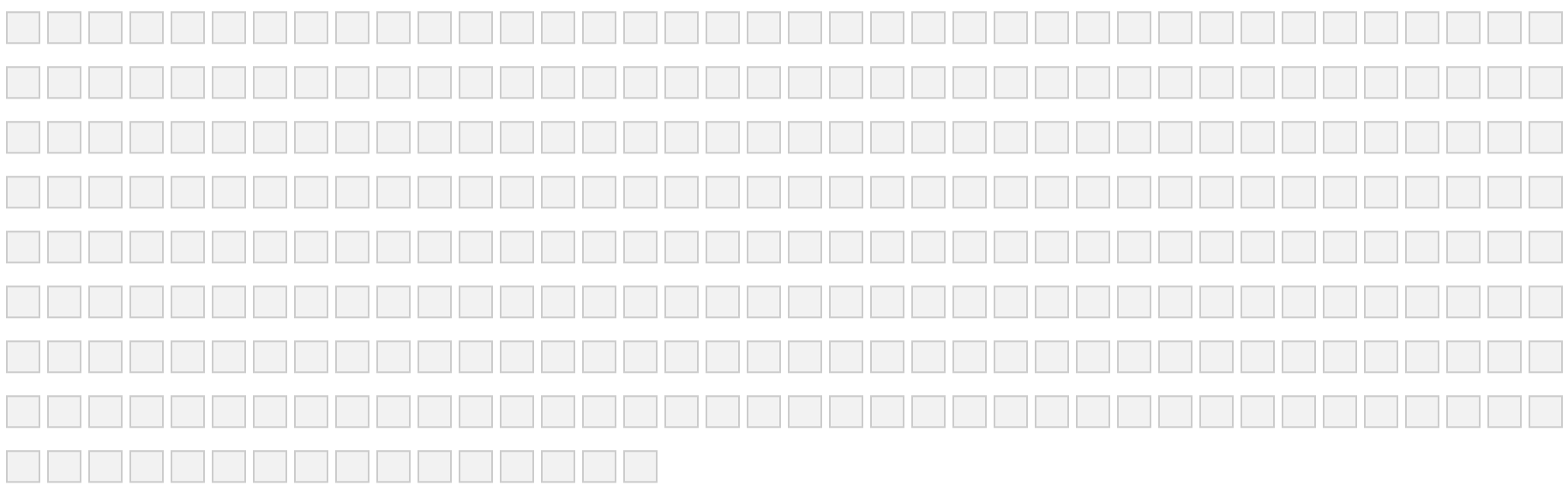




7 Seg 9 Seg

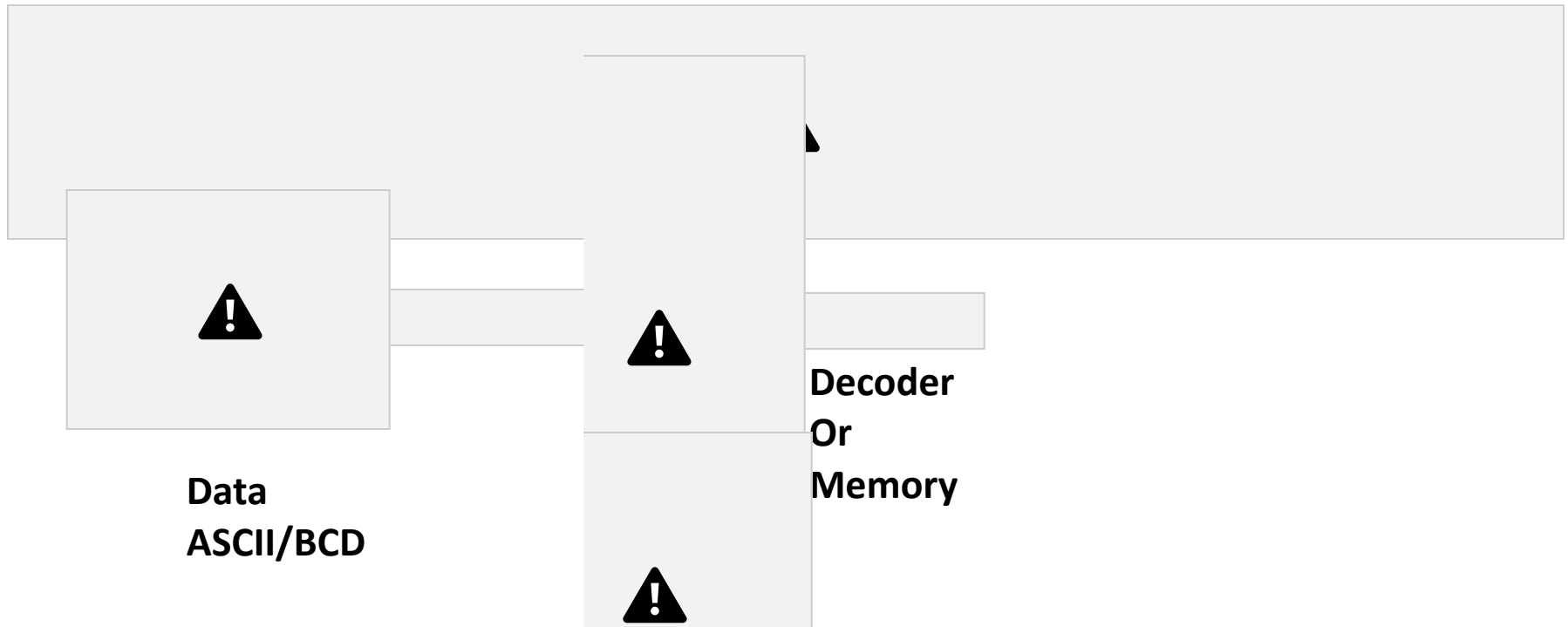
16 Seg 3x5 DotMatix 5x7 9x11





Dot Matrix Display Panel

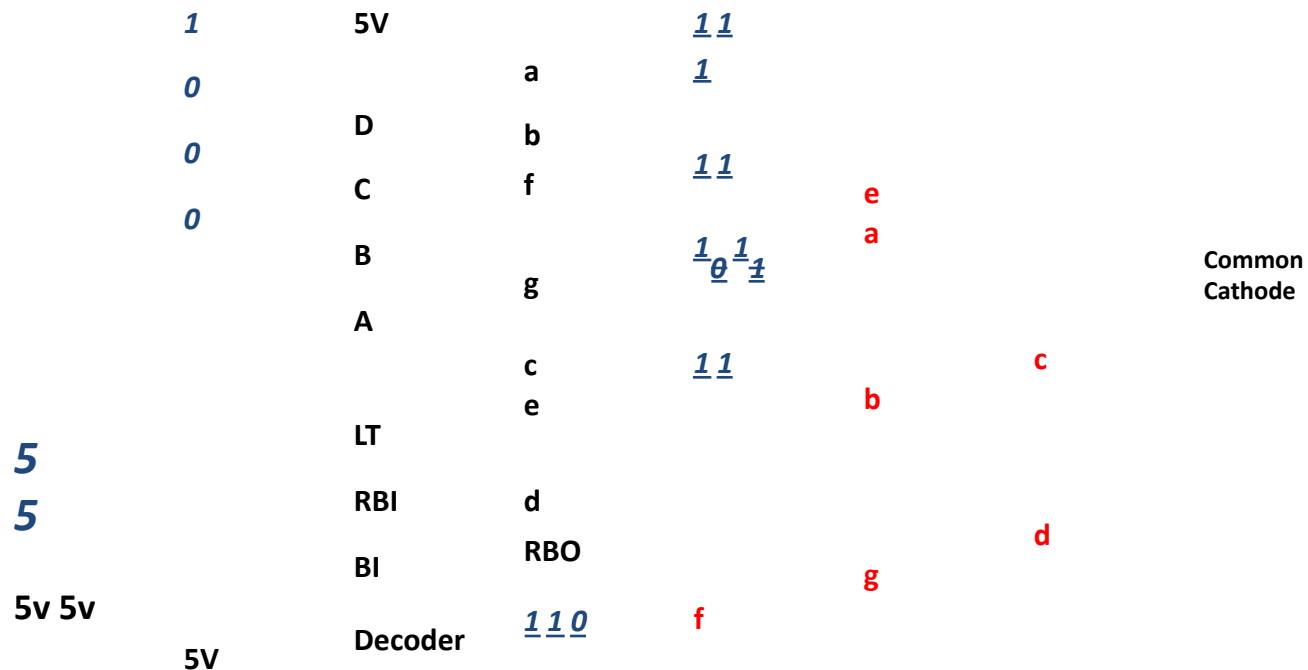
25x80 character monitor



Time to Decode Time to Display



- Data to 7 Segment Decoder





- Data to 7 Segment Decoder

Blank

abc defg
Blank

abc defg

abc defg

5V
abc defg

RBI RBO DCBA

RBI RBO DCBA

abc defg

01

0

RBI RBO DCBA

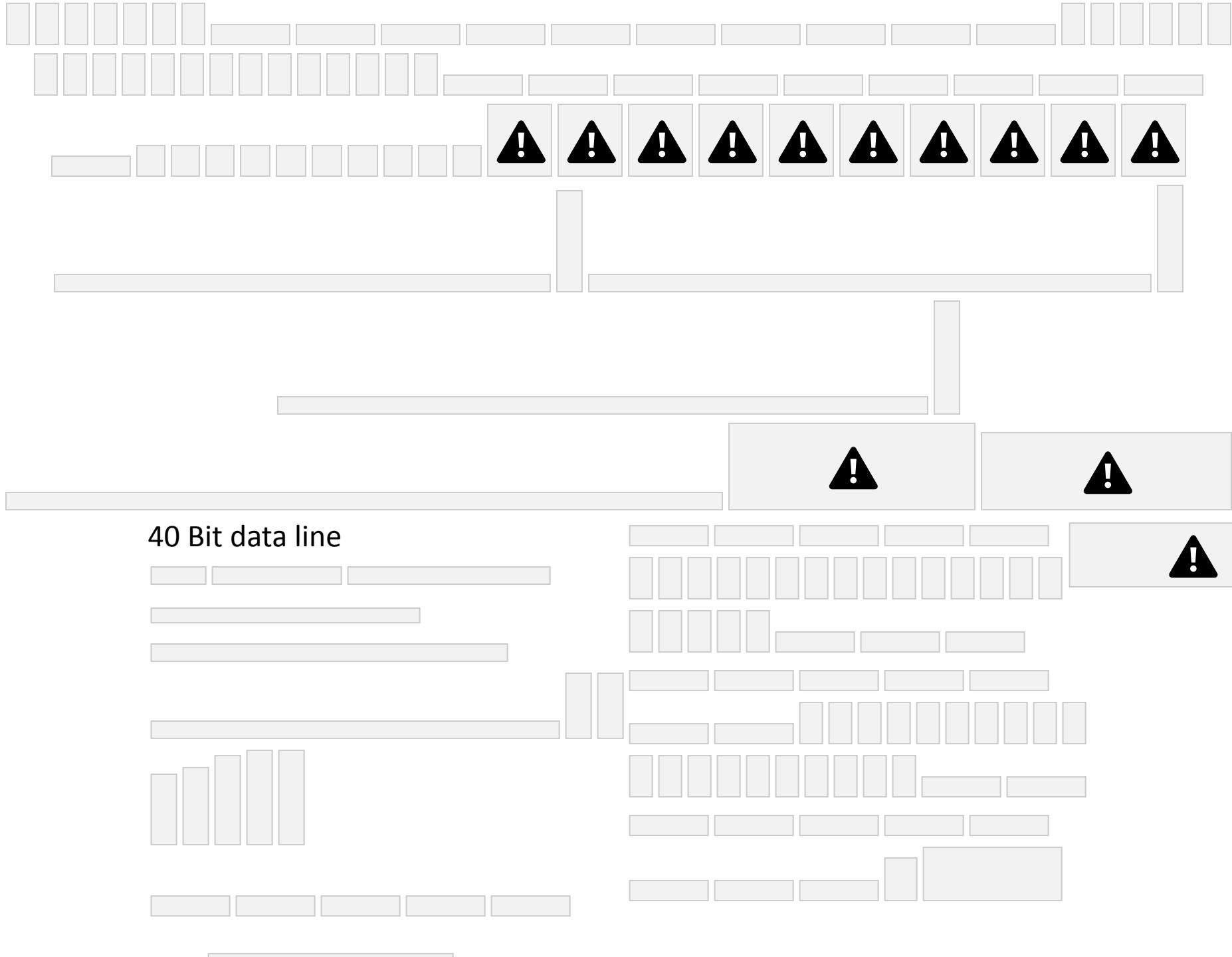
RBI RBO DCBA

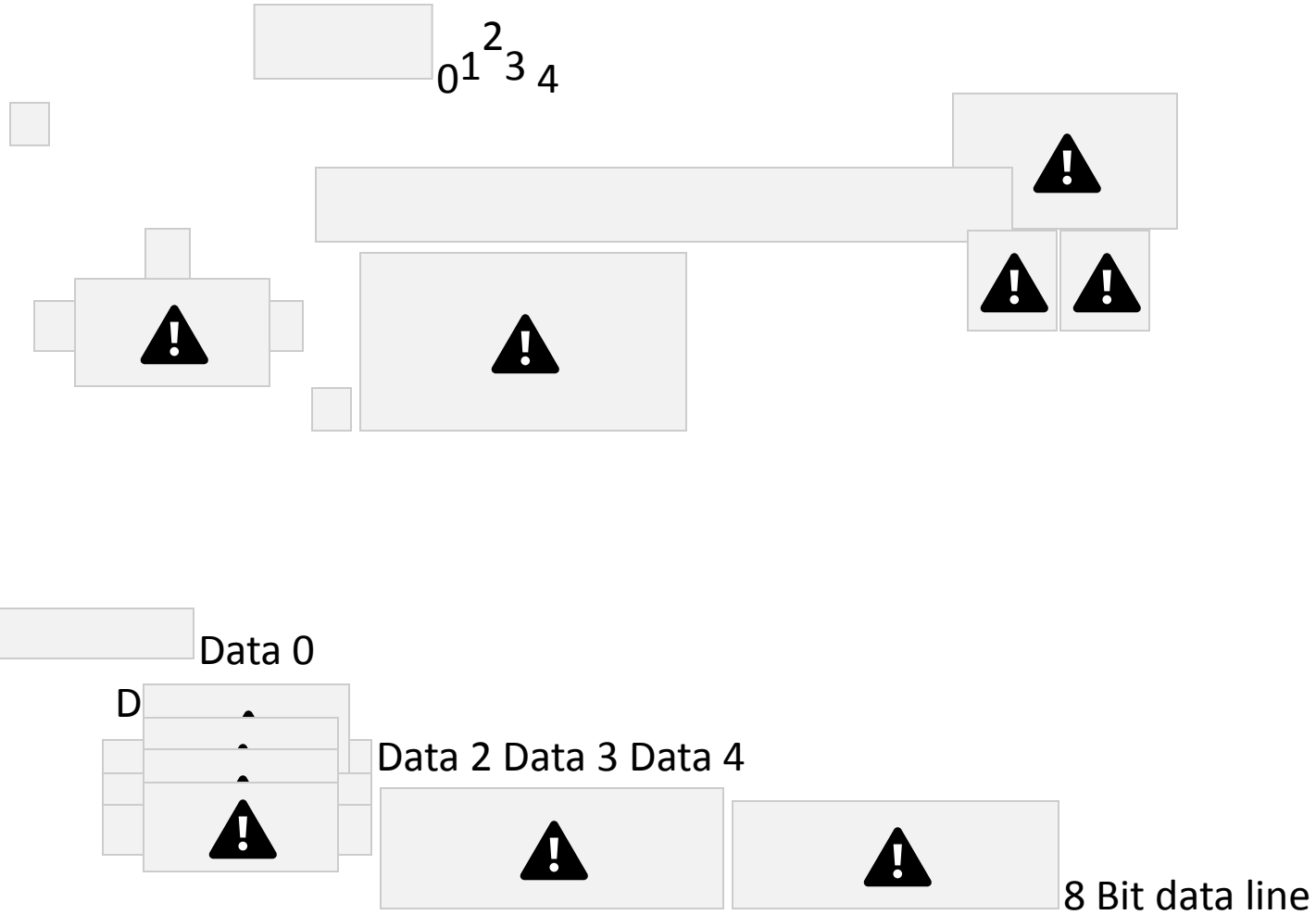
11 RBI RBO

DCBA

00000000001100000111







RAM

590

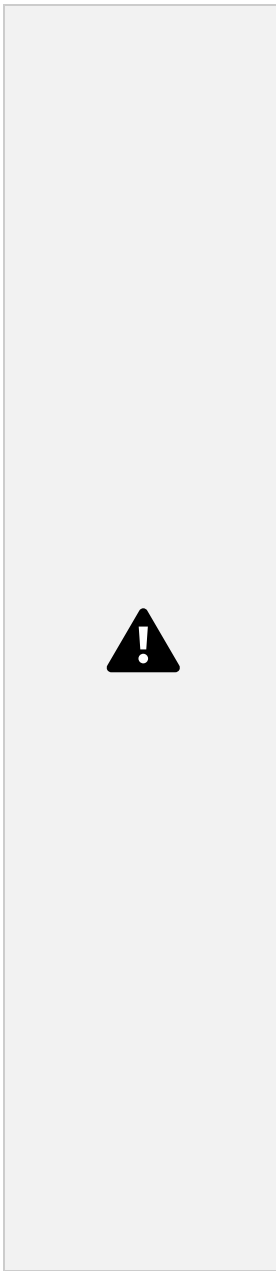
Counter





- Write a program to display contents of B,C,D and E register to LED display
- Write to buffer memory to display to 7 Seg. LEDs
- Use address FC H.FD H.FE H and FF H to display

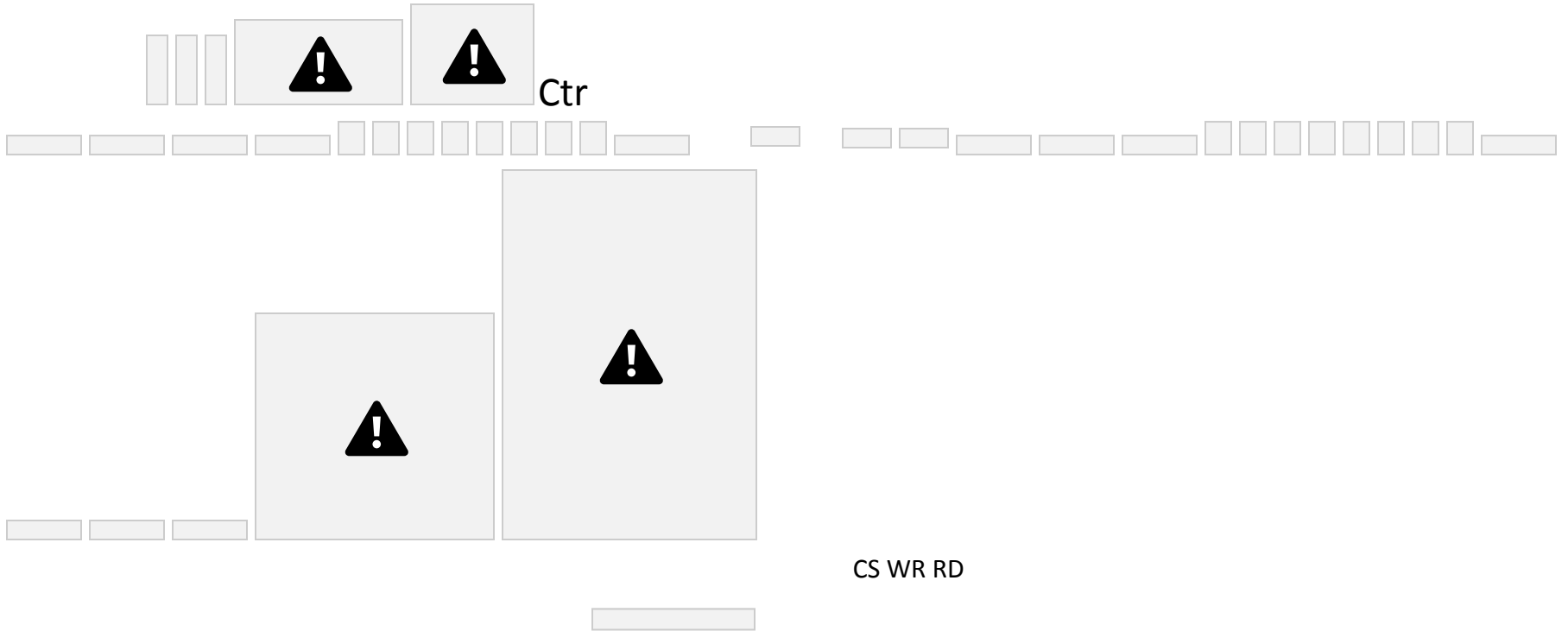




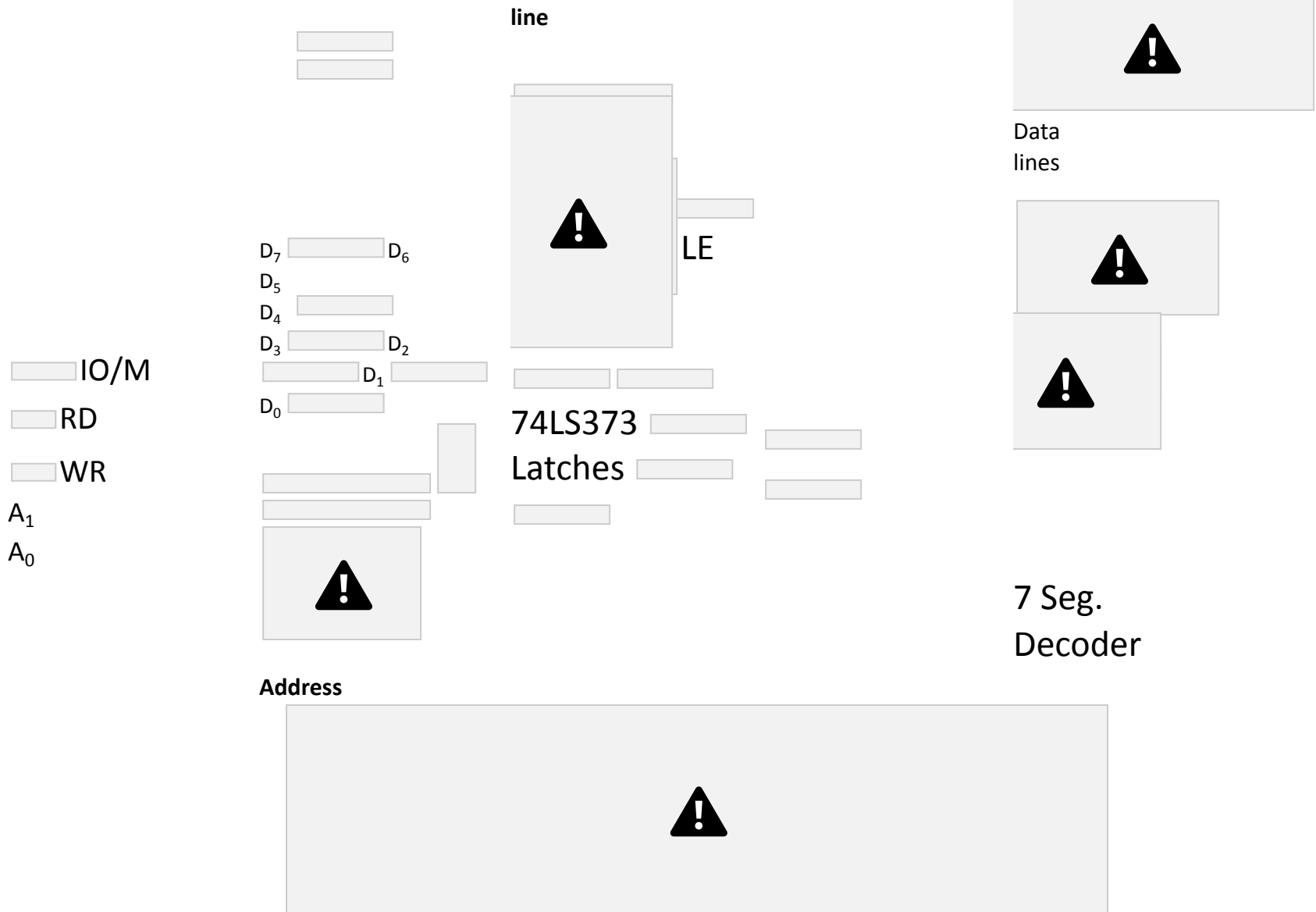
A_7

A_6

A₅
A₄
A₃
A₂



MPU



- Interface program to Display content of B,C,D and E after 1 Sec Delay

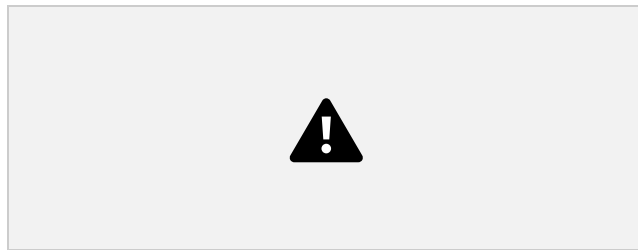
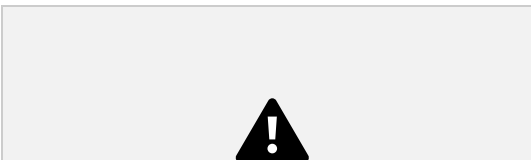
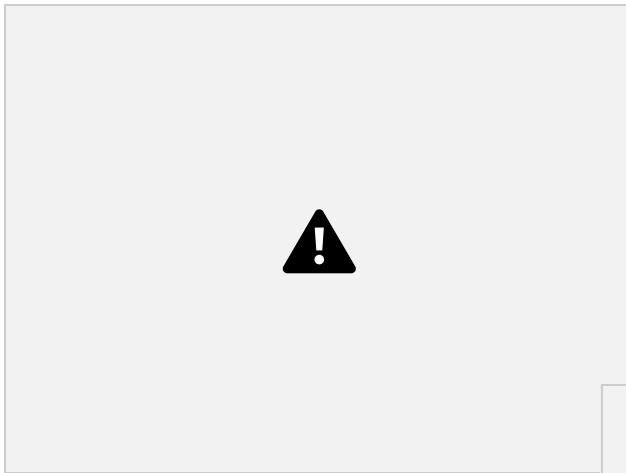
OUT
FCH
MVI
A,C
OUT
FDH
MVI
A,D

DIS: MVI A,B



OUT FEH
MVI A,E
OUT FFH



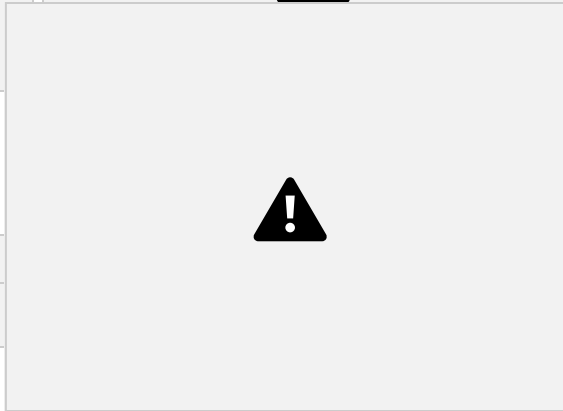
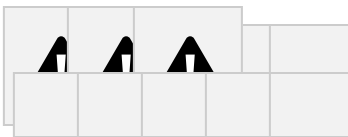


0

1

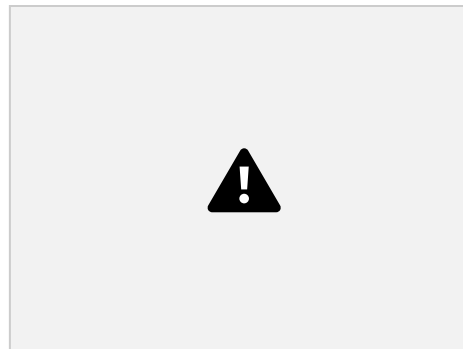
2

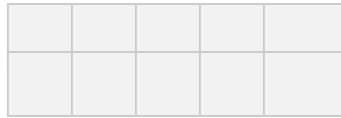
3 4 78 79 0



1

2





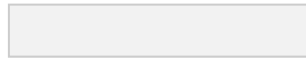
23

24



ROM

Decoder Or



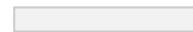
Decoded
Bits

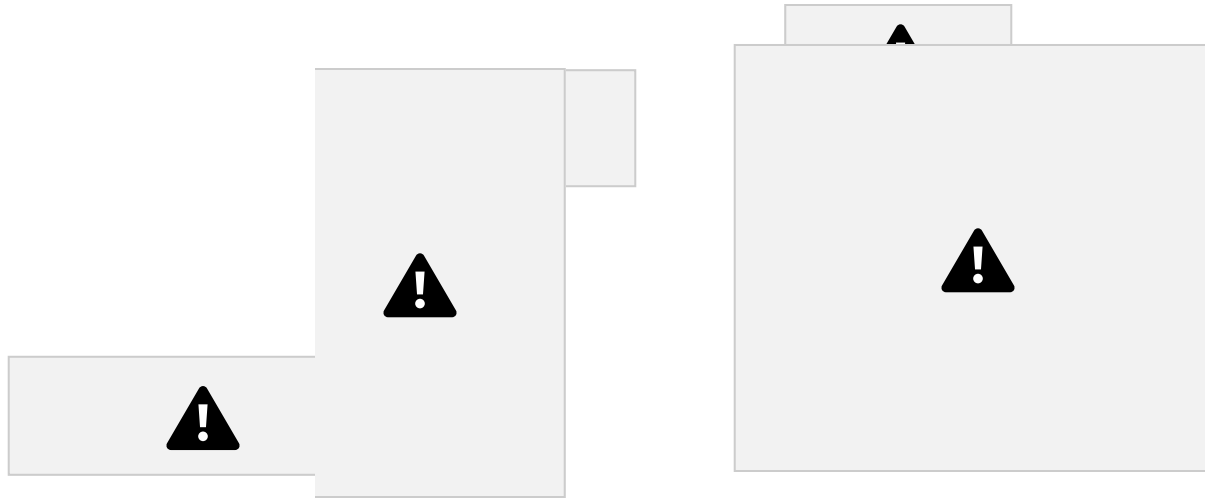


When
Ready
 $C=0$



CS WR





Address (r,c)

Address
line



Data
lines

- Delay of Monitor is 1/50 Sec, After Every 50 Second it makes ready
- From stream buffer it writes to Frame buffer after getting command. (Flush stream command)
- LED is very high speed



– You can write

to LEDs at very high speed – Display meant for to see some thing (Is it good?)

- 30 Frames/Sec is more then enough for human eye (>30 blink make human eye Fixed) • No of blink > 30Hz
 - Human can not differentiate ON/OFF state, it will be seen AS ON state
- Video display make default 30-60 Frame/Sec
- Monitor run at speed 30-60 Frame/Sec



```
for (i=0;i<10000;i++) {  
    printf("HelloWorld");  
}
```

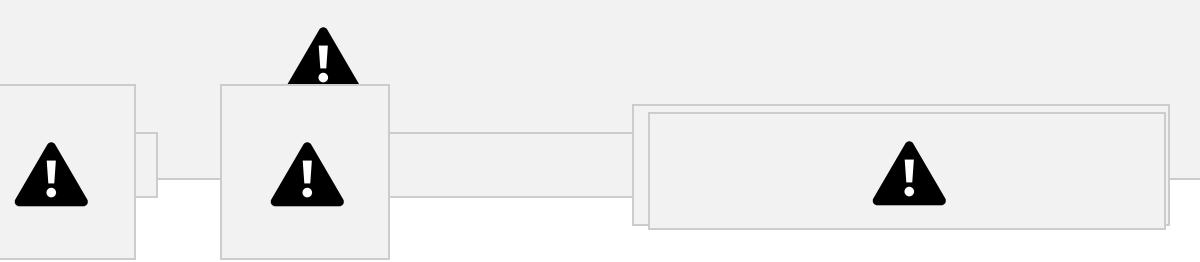
- \$ a.out

- `$ a.out > outputfile`
- `$ a.out > /dev/null`

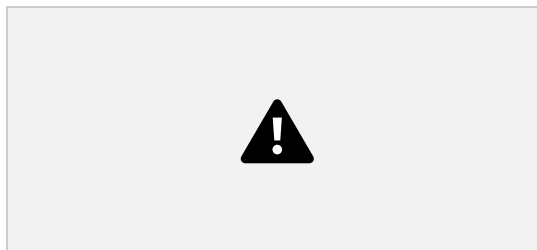
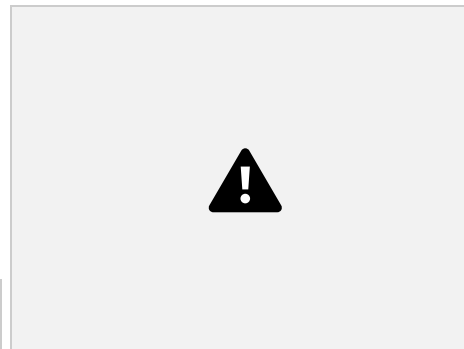
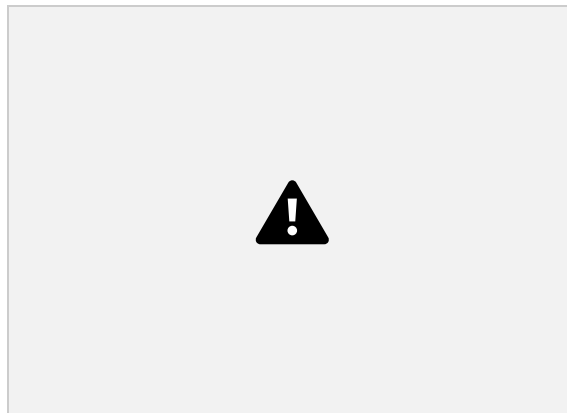


- Char display (80x25 char, 5x7pixel=400x175) • CRT Monitor (400x600, 640x480, 600x800) • LCD Monitor (1024x768, 1280x1024, ...)
- Graphics visually more appealing
- Display Line, Circle, Rectangle, using this
Curve, Polygon – Character primitives
- True type font





RED ARROW
Circle





0 1 2 3 4 1023

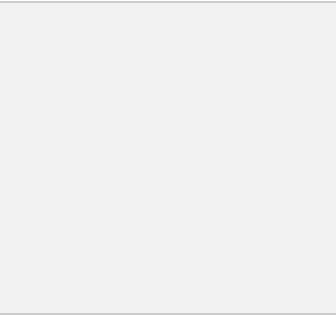
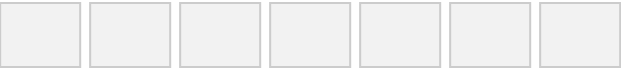


0



1

2



767

1024x768 Pixel LCD

**Frame
Buffer**



8x3=24 Bits

Refresh screen 50 time a Sec



Pixels in Frame
Buffer

1 Bit Per Pixels Pixels on
the Screen

Graphical representation of 1 bit color



15 Bit Per

Pixels in Frame
Buffer

Pixels

Pixels on
the Screen

Graphical representation of 15 bit color



Pixels in Frame
Buffer

24 Bit Per
Pixels

Pixels on



Graphical representation of 24 bit color



1-

bit color 8-bit color 16-bit color



- **Vector graphics:**

- Use geometrical primitives based on mathematical equations, to represent images in computer graphics
- Points, Lines, Curve, shapes or Polygons

- **Raster Graphics:**

Complementary to
Vector Graphics

- Representation of images as an array of pixels



- photographic images
- Store the primitives in the FILE



– Word,

Photoshop, Xfig, GNUPlot, PS, HTML Tag • Raster:
Store the Pixel values in the FILE – BMP, PNG,
JPG

- At runtime, process the primitive and
generate Pixel to raster the screen (Display) •
Who do this?

- Software
- GPU (hardware accelerated routine)

- Computer font formats: TrueType Font



– Where each

letter is created from Bézier curves • Many varieties of font for a character



Units



Points with unit
and curve
properties are
stored

- GPU : specialized processor that

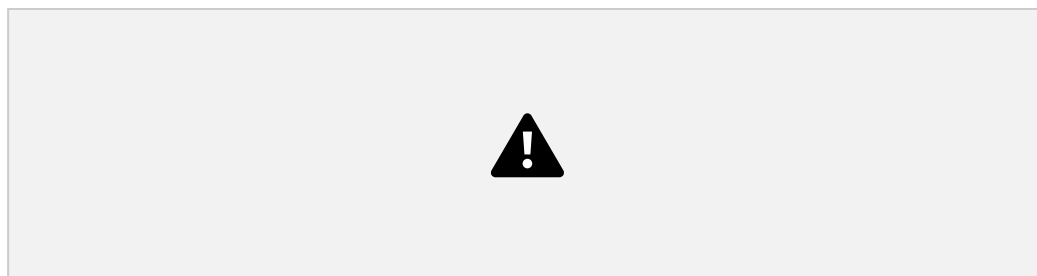
accelerates 3D or 2D graphics primitives operations • Lots of Floating point operations

- Accelerates Primitives

- Line, circle, polygon, mesh, projection, sphere,



3D application



3D API

Commands



3D API:
OpenGL
DirectX/3D

CPU-GPU Boundary

